

MOHAMMED ECH-CHAIBI

Automation & Robotics Engineer

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PROFESSIONAL SUMMARY

I build systems that work — from autonomous robotic harvesters to aerospace helium modelling for a next-generation airship. My background spans PLC programming, machine vision, industrial robotics, and process engineering across three countries and two engineering degrees. I am looking for large-scale, complex automation and controls engineering projects where the work is hard and the impact is real.

[Portfolio: echchaibimohammed.github.io/#about](https://echchaibimohammed.github.io/#about) 

EXPERIENCE HISTORY

Hydroponic Robotic Harvesting System, University of Limerick - Commissioned and integrated an industrial robotic harvesting cell combining a Mitsubishi RV-6S robot, CompactLogix/GuardLogix PLC, FactoryTalk HMI, TCP/IP communication, pneumatics, industrial I/O, and ISO 10218 safety functions. - Programmed the robot in MELFA BASIC IV, implementing Ethernet socket communication, PLC-driven command execution, synchronized robot-PLC handshaking, and autonomous/manual operating modes. - Co-designed and integrated a custom 3D-printed harvesting end-effector with pneumatic actuation, cutting shears, camera mounting, and a Raspberry Pi/OpenCV vision system for lettuce detection and harvest triggering.	12/2025 – 06/2026 Limerick
Rubik's cube Solver, University of Limerick - Programmed an Allen-Bradley PLC to execute a 6-step cube inspection sequence, motor-driven rotation and face imaging. - Designed HMI screens covering operation modes, status monitoring, alarms, and manual override, following GAMP 5 good practice principles. - Implemented IEC 61508-aligned safety functions including E-stop, pause/resume logic, interlocks, and structured alarm management.	02/2026 – 03/2026 Limerick
Bottle Sorting System, University of Limerick - Designed and commissioned a conveyor-based bottle sorting system integrating through-beam and proximity sensors, a pneumatic cylinder, and solenoid valve — achieving consistent and reliable rejection accuracy throughout all test cycles. - Developed LabVIEW/myRIO control logic and built the complete system including conveyor wiring, sensor integration, HMI bottle counting, and Estop functionality.	10/2025 – 12/2025 Limerick
Intern - manufacturing engineer, Flying Whales - Automated helium mass calculations across 50 assembly stages using Excel and Python, reducing verification time from hours to minutes and making the model instantly adaptable to future design changes. - Developed original algorithms to compute helium fill height across 14 independent cells and introduced a systematic imbalance-detection method, replacing manual mass comparison with a threshold-based check against a 2-tonne buoyancy target per cell. - Analysed factory temperature and pressure variations on helium volume behaviour and initiated a risk assessment, identifying key failure scenarios and producing engineering cautions to strengthen the robustness of the filling sequence.	04/2025 – 09/2025 Bordeaux

Intern -Automation and Process Improvement, SITI TEA

05/2024 – 09/2024

Marrakech

- Performed PID tuning on an 8-balance dosing system handling multiple tea product variants, reducing material waste from up to 7% down to 2–5% on the most complex heterogeneous materials.
- Identified material loss at the dosing table through Lean analysis and proposed a vibration and corner recovery system — recommendation formally accepted by management and flagged for future production projects.
- Developed standardised work instructions and proposed layout optimisations to eliminate non-value-added activities and simplify operator workflow.

EDUCATION**Master of engineering: Mechatronics, UNIVERSITY OF LIMERICK**

09/2025 – 08/2026

LIMERICK-Ireland

- *Robotics, automation, control systems, embedded systems, machine vision — with practical work in industrial robotics, PLCs, sensors, and system integration.*

Engineering Degree: Industrial Engineering, IMT NORD EUROPE

09/2023 – 08/2025

Lille-France

- *Industry 4.0 systems thinking, process optimisation, automation, production flow analysis and project management.*

CPGE: Engineering, Ibn Timiya

09/2021 – 09/2023

Marrakech-Morocco

- *Two-year intensive preparatory programme — advanced mathematics, physics, and engineering science, equivalent to first two years of an engineering degree.*

SKILLS**Automation & PLC**

Allen-Bradley Studio 5000, Siemens TIA Portal, State machines & structured sequence design, Emergency stop logic, safety interlocks, Sensor/actuator control&wiring basics.

Software&Tools

Python, C/C++,sensor integration, MATLAB, Git, Serial TCP communication.

Robotics

Mitsubishi RT Toolbox/Melfa Basic, Robot calibration, Robot-PLC communication, 3D Modeling.

Industrial engineering

Lean principles, process analysis, KPI dashboarding, workflow optimisation.

LANGUAGES**French**

Fluent

English

Fluent

Spanish

Elementary

HOBBIES AND INTERESTS

- Chess

- Cooking

- Tennis

CERTIFICATIONS

- CSWA in 3D Modelling

- Lean Six Sigma Green Belt

- MOOC in Project Management.